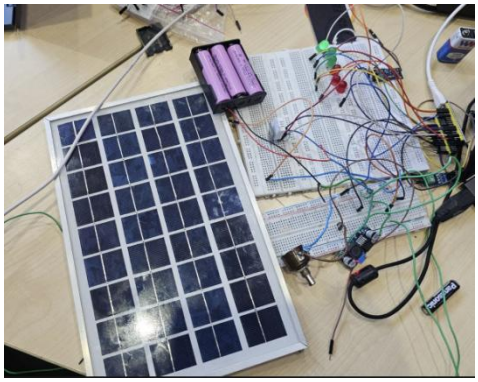
Team 8: EnerGnosis

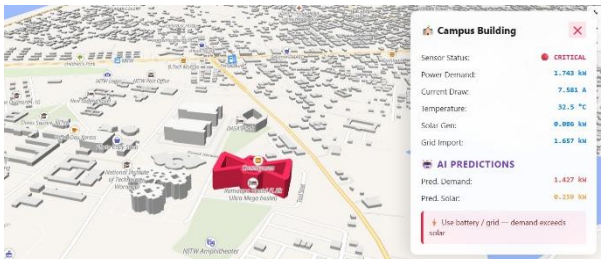


Team members

Hardware

Description: The prototype utilizes an **ESP32-based edge node** to monitor solar harvest, battery health, and ambient conditions via a multi-sensor array (Current, Voltage, Temp, and Light). Powered by a **4S Li-Ion system** with integrated power management, the node transmits processed energy data every 15 seconds to a **Flask-Streamlit stack** for live visualization. To ensure demo reliability, the architecture includes an **IoT simulator fallback** that mirrors hardware behavior when physical nodes are offline.

Software



Built on a modular Python stack, the system utilizes a **Flask API** and **SQLite** backend to unify live ESP32 inputs and simulated data. Real-time ingestion is handled via a custom serial reader with an integrated IoT fallback simulator. The frontend features a **Streamlit** dashboard for analytics and a **MapLibre** 3D digital twin for energy visualization. Finally, a **scikit-learn** layer provides predictive insights for energy demand and solar generation, ensuring a scalable, hardware-agnostic architecture.